

SHAPAI, Kazakhstan

WMO: 353020

Lat: 50.200N

Lon: 51.167E

Elev: 17

StdP: 101.12

Time Zone: 5.00 (E05)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.4	-23.4	-29.5	0.2	-25.9	-26.6	0.3	-23.1	11.6	-6.9	10.2	-6.9	2.0	50	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	36.7	20.1	34.9	19.5	33.1	18.8	21.6	32.1	20.6	31.5	19.8	30.7	3.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.1	25.5	17.0	12.2	24.6	15.8	11.3	24.3	62.6	32.0	59.3	31.3	56.6	30.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.8	7.8	-29.0	39.1	3.7	1.6	-31.6	40.3	-33.8	41.2	-35.9	42.1	-38.6	43.3		
(5)			-28.8	23.0	3.6	1.7	-31.4	24.2	-33.5	25.1	-35.6	26.1	-38.2	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-10.0	-9.3	-1.3	9.5	17.0	22.0	24.7	23.4	16.0	7.6	-0.6	-6.7		
	DBStd	13.54	7.11	6.92	5.74	4.92	4.79	4.62	3.48	4.26	4.53	4.67	5.98	6.43		
	HDD10.0	2529	620	539	352	67	4	0	0	0	6	104	319	517		
	HDD18.3	4414	878	773	610	267	86	17	1	8	97	332	569	775		
	CDD10.0	1721	0	0	1	52	220	361	454	416	185	31	1	0		
	CDD18.3	563	0	0	0	2	43	128	197	167	26	1	0	0		
	CDH23.3	7011	0	0	0	29	528	1607	2414	2066	359	8	0	0		
	CDH26.7	3346	0	0	0	3	196	778	1196	1043	128	2	0	0		
(14) Wind	WSAvg	3.4	3.8	4.0	3.9	3.7	3.3	3.0	2.9	2.9	2.8	3.2	3.4	3.6		
(15) Precipitation	PrecAvg	259	18	14	18	22	24	32	21	17	21	29	23	18		
	PrecMax	439	48	53	41	83	73	87	74	102	83	62	50	45		
	PrecMin	132	7	1	0	1	2	1	0	0	0	1	2	6		
	PrecStd	82	11	10	12	17	16	23	19	22	22	16	14	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	3.2	15.7	26.8	33.4	38.2	38.8	38.6	34.0	24.1	13.5	5.8		
		MCWB	2.4	1.7	9.5	15.4	17.8	20.9	20.5	20.3	17.9	13.8	10.0	4.3		
	2%	DB	1.6	1.8	11.4	23.5	31.1	36.0	36.3	36.5	30.1	21.2	10.2	3.4		
		MCWB	0.8	0.7	6.8	14.0	17.5	19.9	19.8	20.0	16.7	12.6	7.6	2.2		
	5%	DB	0.8	1.0	8.3	21.0	29.0	33.5	34.6	34.5	27.4	18.3	8.3	1.7		
		MCWB	0.1	0.1	5.2	13.0	17.0	18.8	19.5	19.4	15.8	11.2	6.4	0.6		
	10%	DB	-0.1	0.2	5.8	18.4	26.3	31.3	32.6	32.4	24.8	15.6	6.5	0.8		
		MCWB	-0.7	-0.6	3.2	11.5	15.9	18.5	19.1	18.8	14.9	9.9	5.0	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.1	9.8	16.3	19.6	25.1	22.6	22.7	19.4	15.8	11.0	5.0		
		MCDB	3.0	2.8	14.4	24.4	29.1	34.2	33.0	32.9	30.1	21.2	13.2	5.9		
	2%	WB	1.0	1.0	7.3	14.7	18.4	21.6	21.4	21.1	17.7	13.5	8.3	2.3		
		MCDB	1.4	1.5	11.1	22.2	28.6	31.2	32.1	32.5	27.2	19.2	9.7	3.2		
	5%	WB	0.2	0.2	5.3	13.5	17.5	19.9	20.6	20.2	16.6	11.9	6.6	0.9		
		MCDB	0.7	0.8	8.1	20.2	27.3	30.9	31.2	31.9	25.4	17.0	8.1	1.5		
	10%	WB	-0.8	-0.6	3.5	12.0	16.4	18.9	19.9	19.4	15.5	10.5	5.1	0.1		
		MCDB	-0.1	0.2	5.6	17.7	25.5	29.5	30.3	30.8	23.6	14.8	6.5	0.8		
(35) Mean Daily Temperature Range	MDBR		6.9	7.6	7.8	11.3	12.1	12.7	12.5	13.3	12.9	10.1	6.3	5.8		
	5% DB	MCDBR	4.6	5.7	11.3	14.6	15.0	15.4	14.8	15.8	17.1	14.9	8.3	4.5		
		MCWBR	4.5	5.2	7.3	7.2	5.6	5.1	4.7	5.3	7.3	8.3	6.1	4.0		
	5% WB	MCDBR	4.9	5.6	10.4	13.7	14.0	13.6	13.0	14.3	15.0	13.6	7.4	4.0		
		MCWBR	4.9	5.4	7.1	7.3	6.0	5.6	5.2	5.6	7.4	8.4	6.0	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.342	0.394	0.446	0.425	0.412	0.416	0.428	0.382	0.362	0.340	0.319		
	taud		2.063	2.017	2.055	2.050	2.191	2.296	2.292	2.229	2.319	2.298	2.241	2.128		
	Ebn at Noon		666	759	790	793	833	848	837	804	805	745	656	611		
	Edh at Noon		86	116	135	152	138	125	124	126	104	88	73	70		
(44) All-Sky Solar Radiation	RadAvg		1.09	2.23	3.37	4.64	6.07	6.64	6.43	5.54	3.95	2.26	1.04	0.72		
	RadStd		0.27	0.32	0.51	0.35	0.44	0.52	0.40	0.35	0.39	0.27	0.13	0.17		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page